

Multiple Choice (10 marks)

1. A surgical support can be used for:
 - (a) retraction of the stoma.
 - (b) high output stomas.
 - (c) herniation around the stoma.
 - (d) infection around the stoma.
 - (e) dehydration caused by the stoma.
2. All other things being equal, which of the following persons is likely to live the longest?
 - (a) A black woman
 - (b) A Hispanic man
 - (c) A white woman
 - (d) An Asian man
 - (e) A white man
3. What must you ensure before a speaking valve is used?
 - (a) The patient has no difficulty in breathing.
 - (b) It is 24 hours after the tracheostomy tube was first inserted.
 - (c) The patient is able to speak clearly.
 - (d) The patient's vital signs are stable.
 - (e) That the patient is not on a ventilator.
4. The energy released from the breakdown of the high-energy phosphates, ATP and phosphocreatine, can sustain maximal exertion exercise for about:
 - (a) 20-30 minutes.
 - (b) 1-2 seconds.
 - (c) 10-15 minutes.
 - (d) 30-40 seconds.
 - (e) 90-100 seconds.
5. Which of the following is not a beneficial effect of exercise in people with diabetes:
 - (a) Decreased muscle mass
 - (b) Impaired digestion
 - (c) Worsening of vision
 - (d) Increase in blood sugar levels
 - (e) Reduction of bone density

True / False (10 marks)

Answer True or False

6. True or False: The descending loop of Henle is permeable to H^+ ions, allowing their reabsorption in the nephron.
7. True or False: Cognitive symptoms are less likely to be prominent in Parkinson's disease compared to other types of dementia.
8. True or False: It is commonly believed that having fewer children is associated with a longer lifespan in women.
9. True or False: Vitamin C acts as a cofactor for hydroxylation reactions critical for collagen synthesis in connective tissues.
10. True or False: A baby born with pulmonary hypoplasia secondary to oligohydramnios due to renal agenesis is classified as having an association.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the role of ectomesenchyme in the development of melanocytes, and analyze its significance in the broader context of health sciences and tissue differentiation.
12. Discuss the implications of the increasing age at which individuals can receive full social security benefits on public health and economic stability.

End of Paper